INCH-POUND

MIL-DTL-52525/1C
12 November 2008
SUPERSEDING
MIL-DTL-52525/1B
30 June 1998

DETAIL SPECIFICATION SHEET

FITTINGS, HOSE, SCREW-ON TO 37° FLARE FEMALE SWIVEL, STRAIGHT, SAE-J516 TYPE, FOR ASSEMBLY WITH SAE-J517, TYPE 100R2 HOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-52525.

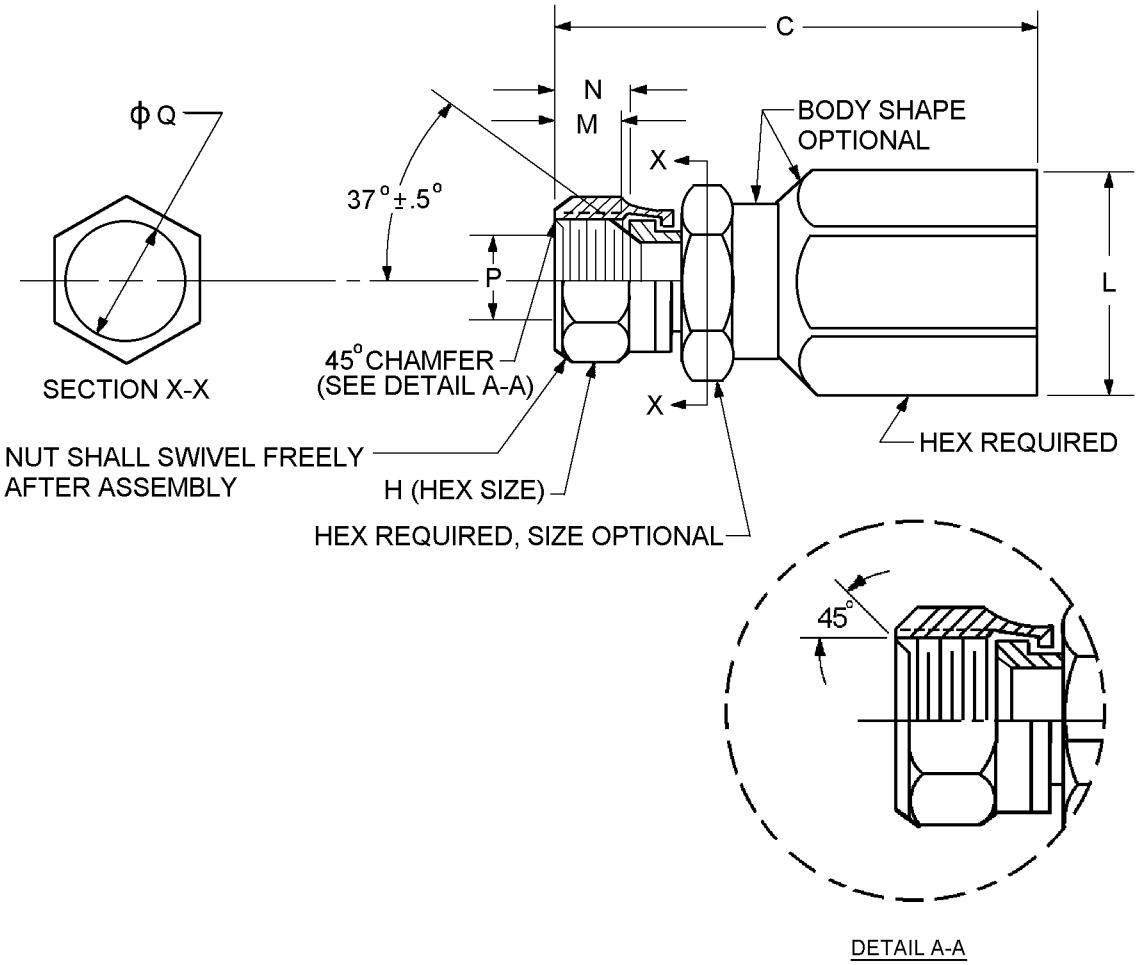


FIGURE 1. Fitting, screw-on to 37° flare female swivel, straight.

AMSC N/A

FSC 4730

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Part or Identifying Number (PIN)	Hose		Tube		Thread size (see note 3)	C max inches (mm)	H Hex size Inches (mm)
	dash no.	ID inches (mm)	dash no.	OD inches (mm)			
M52525/1-4-4	4	.250 (6.35)	4	.250 (6.35)	.438-20-UNF	2.88 (73.15)	.563 (14.30)
M52525/1-4-5	4	.250 (6.35)	5	.313 (7.95)	.500-20-UNF	2.94 (74.68)	.625 (15.88)
M52525/1-6-6	6	.375 (9.53)	6	.375 (9.53)	.563-18-UNF	3.16 (80.26)	.688 (17.48)
M52525/1-8-8	8	.500 (12.70)	8	.500 (12.70)	.750-16-UNF	3.33 (84.58)	.875 (22.23)
M52525/1-8-10	8	.500 (12.70)	10	.625 (15.88)	.875-14-UNF	3.38 (85.85)	1.000 (25.40)
M52525/1-12-12	12	.750 (19.05)	12	.750 (19.05)	1.063-12-UNC	4.08 (103.63)	1.250 (31.75)
M52525/1-16-16	16	1.000 (25.40)	16	1.000 (25.40)	1.313-12-UNC	4.54 (115.32)	1.500 (38.10)
M52525/1-20-20	20	1.250 (31.75)	20	1.250 (31.75)	1.625-12-UNC	5.47 (138.94)	2.000 (50.80)
M52525/1-24-24	24	1.500 (38.10)	24	1.500 (38.10)	1.875-12-UNC	5.85 (148.59)	2.250 (57.15)
M52525/1-32-32	32	2.000 (50.80)	32	2.000 (50.80)	2.500-12-UNC	6.53 (165.86)	2.875 (73.03)

PIN	L max inches (mm)	M min inches (mm)	N inches (mm) +.030 (0.76) -.015 (0.38)	P (see note 4) inches (mm) ±.005 (0.13)	Q min bore dia. inches (mm)
M52525/1-4-4	.89 (22.61)	.300 (7.62)	.344 (8.74)	.295 (7.49)	.11 (2.79)
M52525/1-4-5	1.00 (25.40)	.331 (8.41)	.375 (9.53)	.355 (9.02)	.11 (2.79)
M52525/1-6-6	1.19 (30.23)	.333 (8.46)	.375 (9.53)	.435 (11.05)	.26 (6.60)
M52525/1-8-8	1.24 (31.50)	.375 (9.53)	.422 (10.72)	.570 (14.48)	.375 (9.53)
M52525/1-8-1	1.24 (31.50)	.461 (11.71)	.500 (12.70)	.680 (17.27)	.375 (9.53)
M52525/1-12-12	1.75 (44.45)	.469 (11.91)	.562 (14.27)	.850 (21.59)	.57 (14.48)
M52525/1-16-16	2.05 (52.07)	.563 (14.30)	.594 (15.09)	1.100 (27.94)	.78 (19.81)
M52525/1-20-20	2.63 (66.80)	.563 (14.30)	.625 (15.88)	1.410 (35.81)	1.01 (25.65)
M52525/1-24-24	2.94 (74.68)	.661 (16.79)	.734 (18.64)	1.625 (41.28)	1.25 (31.75)
M52525/1-32-32	3.50 (88.90)	.844 (21.44)	.938 (23.83)	2.240 (56.90)	1.72 (43.69)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Threads shall be in accordance with ASME B1.1 series with graded pitches fine (UNF) and series with constant pitches (UNC), class 2B.
4. Dimension P to be concentric with dimension Q with in .015 inch (0.38 mm) full indicator movement.
5. All dimensions not shown shall be in accordance with SAE-J516.
6. This is similar to SAE-J516 type for use with SAE-J517, type 100R2 hose

FIGURE 1. Fitting, screw-on to 37° flare female swivel, straight - Continued.

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REQUIREMENTS:

Fittings shall be as specified on figure 1 and in tables I, II, and III.

The fittings described herein are for use with MIL-DTL-52471/5, SAE-J517, type 100R2 hose, which is high pressure double steel wire braid reinforced and MIL-DTL-52471/2 hose assemblies.

If fittings are to be used on an oxygen hose, see MIL-DTL-52525.

Materials and finishes shall be in accordance with MIL-DTL-52525 and in table I. All finishes shall be capable of withstanding 96 hours minimum of salt spray.

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TABLE I. Material and finish identification codes. 1/ 2/

PIN code material/plating finish	Material	Plating finish
A	Aluminum alloy 6061 or 7075	Anodize in accordance with MIL-A-8625, type II.
AN		Anodize in accordance with MIL-A-8625, type II and NAVAIR trivalent chromium pretreatment (TCP) in accordance with MIL-DTL-81706, type 2, class A.
BN		Bare aluminum with NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
C	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2.
CN		Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 and NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
YC	Chrome-molybdenum steel alloy 4130	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2.
YN		Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 and NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
F	Steel	NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
FN	Chrome-molybdenum steel alloy 4130	
G	Steel	Zinc plating with colorless passivate in accordance with ASTM B633, type V, Fe/Zn 25.
YG	Chrome-molybdenum steel alloy 4130	
H	Steel	Zinc phosphate finish in accordance MIL-DTL-16232, type Z, class1. 3/
YH	Chrome-molybdenum steel alloy 4130	
J	Steel	Zinc plating with chromate conversion in accordance with ASTM B633; type II or III, Fe/Zn 5, or ASTM B695, type II, class 5.
YJ	Chrome-molybdenum steel alloy 4130	
K	Nickel-copper alloy UNS N04400	No additional finish.
M	High-chromium nickel alloy UNS N06690	No additional finish.
N	Steel	Zinc aluminum in accordance with ASTM F1136, grade 3, NC.
YN	Chrome-molybdenum steel alloy 4130	
P	Steel	Zinc plating with colorless passivate in accordance with ASTM B633, type VI, Fe/Zn 5.
YP	Chrome-molybdenum steel alloy 4130	
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, type 6 or 7.
SN		Passivation above and NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
T	Titanium	Anodize in accordance with SAE-AMS2488, type 2.
V	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 1.
W	Steel	Any zinc plating above.
YW	Chrome-molybdenum steel alloy 4130	
Z	Steel	Any zinc plating above with NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
YZ	Chrome-molybdenum steel alloy 4130	

1/ All materials and finishes shall be in accordance with MIL-DTL-52525.
2/ Embitterment test need not be run.
3/ Zinc phosphate finish is hexavalent chromium free.

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Maximum operating pressure shall be as specified in table II. Maximum operating pressures are for low carbon steel fittings consult, manufacturer for values on other materials.

TABLE II. Maximum operating pressures of fittings. 1/ 2/

SAE dash size	Hose ID inches (mm)	psi	MPa
-4	.250 (6.35)	4500	31
-6	.375 (9.53)	4000	27.5
-8	.500 (12.70)	4000	27.5
-12	.750 (19.05)	3000	21
-16	1.000 (25.40)	2500	17
-20	1.250 (31.75)	2000	14
-24	1.500 (38.10)	1500	10.5
-32	2.000 (50.80)	1125	8

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

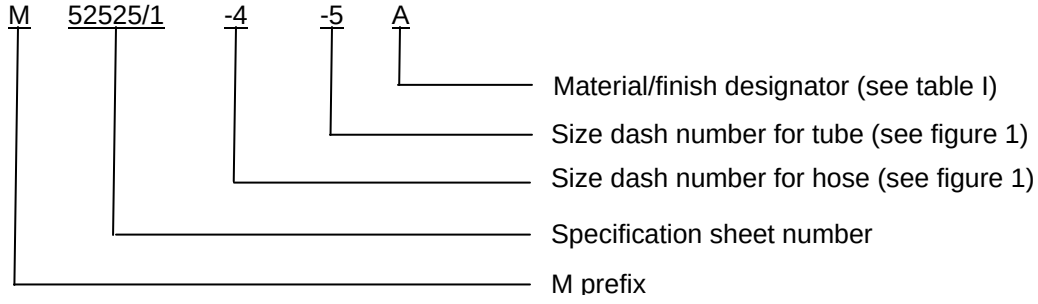
Over tightening torque shall be as specified in table III. Over tightening torque values are for low carbon steel fittings, consult manufacturer for values on other materials.

TABLE III. Over-tightening torque for swivel nut. 1/ 2/ 3/

SAE dash size	Hose ID ref inches (mm)	Torque	
		Pound-force Inches (lb-in)	N m
-4	.250 (6.35)	135	15
-6	.375 (9.53)	270	31
-8	.500 (12.70)	450	51
-12	.750 (19.05)	900	102
-16	1.000 (25.40)	1200	136
-20	1.250 (31.75)	1900	215
-24	1.500 (38.10)	2700	305
-32	2.000 (50.80)	4000	452

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.
3/ Test torques are for low carbon steel fittings.
Consult manufacturer for values for other materials.

PIN: The PIN consists of the letter “M” the specification sheet number, a dash, a number for hose size, a dash, a number for tube size, and a material finish designator.



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PIN example: M52525/1-4-5A, describes a connector for a .250 inch hose OD, and a .313 inch tube OD, aluminum with an anodized finish.

To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when the other materials and finishes specified in this document cannot meet performance requirements.

Color identification. Color identification shall be in accordance with SAE-AS4841.

Marking shall include the manufacturer's name or trademark, size, and hose identifier R2. Location of marking is optional.

Harmonization to SAE-J516 cross reference PIN in table IV is for reference only. SAE style fittings may not have the same material/finish as required by MIL-DTL-52525 and legacy with MIL-DTL-52471.

TABLE IV. Harmonization to SAE-J516 cross reference PIN. 1/

PIN	SAE PIN
M52525/1-4-4	SAE J516 4-4 F240154
M52525/1-4-5	SAE J516 5-4 F240154
M52525/1-6-6	SAE J516 6-6 F240154
M52525/1-8-8	SAE J516 8-8 F240154
M52525/1-8-10	SAE J516 10-8 F240154
M52525/1-12-12	SAE J516 12-12 F240154
M52525/1-16-16	SAE J516 16-16 F240154
M52525/1-20-20	SAE J516 20-20 F240154
M52525/1-24-24	SAE J516 24-24 F240154
M52525/1-32-32	SAE J516 32-32 F240154

1/ Material and finish designators are omitted from PIN's, see table I and SAE-J846.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-52525, this document references the following:

MIL-A-8625
MIL-DTL-16232
MIL-DTL-52471
MIL-DTL-52471/2
MIL-DTL-52471/5
MIL-DTL-81706
ASME B1.1
ASTM B633
ASTM B695
ASTM F1136
SAE-AMS-C-81562
SAE-AMS-QQ-P-416
SAE-AMS2417
SAE-AMS2488
SAE-AMS2700
SAE AS4841
SAE-J516
SAE-J517
SAE-J846

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